Jeremy J. Burkes

Data Analyst | SQL, Python, Tableau, Excel | Optimizing operational workflows and driving efficiency

Email | LinkedIn | Portfolio

PROFESSIONAL SUMMARY:

I've spent 20 years solving other people's problems well enough to keep getting handed bigger ones. From a GIS Analyst who learned SQL, to an operations manager who squeezed a citywide operation out of a small staff, to a guy who builds his own fantasy football spreadsheets for fun. I'm a Data Analyst and GIS professional with a continuous progression of modern technical skills - including SQL, Tableau, Python, and Excel. My technical base combined with my public service experience collaborating with diverse stakeholders has enabled me to act as an effective translator across domains. I'm comfortable working remotely, collaborating with distributed teams, and handling projects from end-to-end. As a previous manager of a large municipal division, I care about data quality, attention to detail, and understand that dashboards are only valuable when the numbers are trusted.

TECHNICAL SKILLS:

- **Languages & Querying:** SQL, Jupyter Notebook, Databricks, Python (Pandas, Numpy)
- **Data Visualization:** Tableau, ArcGIS Online, Excel (Pivot Tables, XLookup)
- **Geospatial & Operations:** ArcGIS Pro, Cityworks, SAP, DBeaver, Jira, Trello

PROFESSIONAL EXPERIENCE:

City of Houston: Parks & Recreation, Houston, TX — Sep 2020 – May 2026

Division Manager

- Managed multi-year vendor contracts and operational workflows for municipal tree maintenance, planting, and watering services, overseeing contractor throughput, service-level compliance, and performance metrics across 4,500+ annual work orders.

- Created and maintained 4 dynamic dashboards in ArcGIS Online to track funding allocation and ROI metrics for tree planting initiatives, directly informing 4 key donor and stakeholder groups.

- Communicated data-informed insights on urban tree management into clear, actionable presentations for civic organizations, HOAs, community groups, government stakeholders, and local media.

City of Houston: Parks & Recreation, Houston, TX — Aug 2014 – Sep 2020

Senior Operations Manager

- Enforced complex municipal code compliance across 2,500+ annual plan reviews, translating regulatory standards into clear operational guidelines for builders and reducing compliance violations by 28% through stakeholder education.

- Optimized tree protection operations, increasing overall efficiency by 81% and dropping residential plan review turnaround times from 36 days down to 7 days.

- Engineered process redesign and mobile cloud integration, eliminating 4.5 hours of daily manual reporting by automating data collection workflows in the field.

- Analyzed 435,000+ street-tree inventory records with SQL and GIS, identifying trends in tree health, growth, insect activity, and disease to drive proactive maintenance decisions.

City of Houston: Parks & Recreation, Houston, TX Aug 2012 – Aug 2014
Operations Manager

- Developed standardized protocols that reduced service level response time for tree inspections by 90%, from 21 days to 2 days.

- Automated bi-weekly Excel reports with pivot tables and charts that highlighted performance measures for tree maintenance operations, earning consistent positive feedback from department leadership for the quality and timeliness of work.

- Managed a $4.6 million multi-year budget for contracted tree maintenance services, responding to over 6,000 customer-reported service requests each fiscal year and handling over 4,500 work orders with 4 contractor crews and 1 in-house crew annually.

City of Houston: Parks & Recreation, Houston, TX Dec 2006 – Aug 2012
Geographical Information Systems Analyst

- Queried and analyzed 20,000+ work order records using SQL and ArcMap, generating spatial heat maps that targeted neighborhoods for proactive maintenance.

- Uncovered critical canopy vulnerability trends through descriptive analytics, forecasting a 40%+ canopy loss over the next 20 years in high-risk communities experiencing simultaneous tree species decline.

- Helped foster shared stewardship opportunities within vulnerable communities by identifying available planting space along major corridors and targeting marketing campaigns to local businesses and stakeholders to solicit participation in the city's tree adoption programs, which saw a 35% year-over-year increase.

- Managed and maintained a growing dataset of over 400,000 public street-tree, stump, and planting-space inventory records, and transformed inventory data into data-driven annual reports supporting community tree management planning for 46 of 88 neighborhoods.

- Performed requirements gathering and data governance for 138 citywide master feature classes, raster images, and sensitive PII location data as a member of the City's Enterprise GIS Technical Committee and Enterprise GIS Governance Committee, enforcing strict version control and mandatory metadata standards to ensure datasets were fit for operational decision-making.

City of Houston: Parks & Recreation, Houston, TX Nov 2005 – Dec 2006
Geographical Information Systems Technician

- Cleaned, normalized, and expanded an active street-tree inventory by 86% (growing from 121,398 to 225,635 records) within 4 months by redesigning data dictionaries and mobile collection processes.

Schoor DePalma: Engineers & Consultants, Manalapan, NJ May 2005 – Aug 2005
Geographical Information Systems Survey Analyst I

- Managed and manipulated geospatial datasets using Microsoft Access, providing critical mapping and data research support to project managers, outside clients, and planners while establishing metadata standards to ensure data traceability and usability.

- Attended weekly meetings with project stakeholders to understand their data needs and goals and documented project requirements and expected workflows for the GIS team.

RECENT ANALYTICS PROJECTS:

SLA Analysis & Service Trends | *SQL*, *Jupyter Notebook, Python, DBeaver, Tableau* | Review Project

- Analyzed public service delivery datasets comprising nearly 39,000 records to establish tired P50-P90 target thresholds, reducing overall SLA breach rates from 56.9% down to 15%. Utilized Pandas ".loc" conditioning, string cleaning, and null value troubleshooting to build analytical pipelines.

Supply Chain Analysis for Fashion & Beauty Startup | *SQL, DBeaver, Python, Tableau* | Review Project

- Evaluated end-to-end inventory and shipping records to quantify defect rates and supplier lead-time variances across product lines, building interactive dashboards to monitor logistical bottlenecks. Identified $10,470 in direct manufacturing cost savings per cycle.

Business Intelligence Dashboard for Urban Forestry | *SQL, DBeaver, Python, Tableau* | Review Project

- Built relational fact and dimension database tables in PostgreSQL and wrote optimized SQL queries to track operational efficiency, maintenance cycle times, and public safety KPIs.

EDUCATION:

Lafayette College, Easton, PA Aug 2000 – May 2004

- Major: Bachelor of Arts in Engineering Studies

United States Military Academy Prep School, West Point Jul 1999 – Jun 2000

- Cadet Candidate E-1 – Bravo Company

CERTIFICATIONS:

- Wildfire Risk Reduction Qualification Mar 2019 – Mar 2028
- Tree Risk Assessment Qualification May 2017 – May 2027
- ISA Certified Urban Forest Professional Jun 2018 – Jun 2028
- ISA Certified Arborist Apr 2013 – Jun 2028